

Hospital Smart Queue System — Presentation Roles

Team of 5 Members

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Split along the module boundaries and system architecture of the enterprise codebase. Each member is assigned a formal engineering role alongside their presentation module ownership, matching their direct code implementation (API controllers, business logic rules, data schema, UI portals, and machine learning models).

Module Ownership & Engineering Roles

5 members

Mg La Yaung Htut

MEMBER 1 • LEAD SYSTEM ENGINEER

ENGINEERING ROLE

Lead System Engineer / System Architect

Patient Module & System Overview

Core system architecture, auth/session control, appointment booking pipeline, and the patient-facing web application. Opens the presentation with the problem statement, enterprise layered architecture, and overall technical vision upon which all other modules integrate.

api/PatientApiController.java

repository/PatientRepository.java

config/WebConfig.java

routes/dashboard.tsx, appointments.tsx

routes/register.tsx, login.tsx

Presents:

Introduction & problem statement, enterprise layered architecture diagram, followed by a live walkthrough of patient sign-up → authentication → doctor booking → queue status tracking.

Ma Hsu Shoon Wutyi

MEMBER 2 • QUEUE ENGINE ENGINEER

ENGINEERING ROLE

Algorithm & Queue Engine Engineer

Smart Queue Engine

The algorithmic backbone of the system: dynamic queue priority ranking, doctor availability detection, emergency priority escalation, and scheduled missed-ticket lifecycle management — housing the core business rules of the hospital.

rule/QueueRule.java

rule/PriorityRule.java

rule/DoctorAvailabilityRule.java

rule/EmergencyRule.java

service/QueueExpiryScheduler.java

Presents:

Algorithmic queue ticket assignment and live priority re-ordering (EMERGENCY > APPOINTMENT > NORMAL), doctor capacity math, concurrency handling, and automated missed-queue expiry background jobs.

Page 1 of 3

Ma Noo Noo Aung

MEMBER 3 • CLINICAL WORKFLOW ENGINEER

ENGINEERING ROLE **Clinical Workflow & Frontend Engineer**

Doctor & Staff Module

The operational portals that directly interact with the live queue: clinical consultation state transitions (calling, in-consultation, pausing, completing) and staff front-desk reception, walk-in registration, doctor re-assignments, and verified emergency triage admissions.

api/DoctorApiController.java

api/StaffApiController.java

service/QueueService.java

routes/doctor.queue.tsx

routes/staff.emergency.tsx

Presents: Live doctor consultation lifecycle (call next patient → complete consultation), staff desk triage intake, doctor reassignments, and front-desk emergency registration dynamically jumping the line.

Ma Ei Pone Chit

MEMBER 4 • DATABASE DESIGNER & DBA

ENGINEERING ROLE **Database Designer & Administrator (DBA)**

Admin Module, Database & Reports

Relational database schema modeling, Flyway migration management (V1 to V11), connection pooling, entity-relationship integrity, and administrative oversight covering hospital departments, doctor profiles, staff accounts, system settings, and analytics reporting.

api/AdminApiController.java

db/migration/V1...V11

config/DatabaseConfig.java

repository/AdminUserRepository.java

routes/admin.reports.tsx

Presents: Hospital ER diagram and database relational schema, Flyway migrations, admin management portal (doctors, departments, users), and generated doctor-performance & queue-flow analytical reports.

Ma Phoo Thet Ngone

MEMBER 5 • AI & MACHINE LEARNING

ENGINEERING ROLE AI / Machine Learning Engineer

AI Features & Conclusion

The machine learning and intelligence layer built atop the enterprise database: predictive patient wait-time estimation, appointment no-show probability scoring, symptom-to-specialty triage classifier, and a hospital knowledge-base RAG chatbot.

api/AiApiController.java

ml/NoShowPredictionService.java

ml/WaitTimePredictionService.java

ml/RagChatbotService.java

Presents: Live AI symptom assessment recommending specialist departments, real-time wait-time and no-show predictions, the interactive RAG hospital assistant, followed by project challenges, lessons learned, and future enhancements.

Suggested Run of Show ~20 min presentation

01	Introduction & Problem Statement — Mg La Yaung Htut	Lead System Engineer	2 min
02	Architecture Overview & Patient Module Demo — Mg La Yaung Htut	System Architect	3 min
03	Smart Queue Engine Walkthrough & Logic — Ma Hsu Shoon Wutyi	Queue Engine Engineer	4 min
04	Doctor & Staff Portals Live Demo — Ma Noo Noo Aung	Clinical Workflow Eng.	4 min
05	Database Schema, ERD & Admin Reports — Ma Ei Pone Chit	Database Designer & DBA	4 min
06	AI Symptoms, Wait-Time & RAG Chatbot Demo — Ma Phoo Thet Ngone	AI / ML Engineer	3 min
07	Project Evaluation, Lessons Learned & Q&A — All Members (Led by Phoo Thet Ngone)	All Team Members	3–5 min
7 Segments Total		Full Team Demo	≈ 20–23 min